

Abhishek Singh

+91 9884177514 | hiabhishek1997here@gmail.com | github | linkedin

PROFESSIONAL HIGHLIGHTS

- 6 years of experience in full-stack development across finance, insurance, retail, telecom, and HR domains.
 - Built production **Agentic AI** systems using **LangChain**, **Google ADK**, **MCP**, **A2A**, And **AG-UI**, backed by RAG, vector search, automated evals, and LLM-native observability.
 - Built REST and **GraphQL** APIs using **Java**, **Spring Boot**, and **NestJS**; handled auth, throttling, caching, and fault tolerance; scaled to handle **10,000+** users.
 - Built front-end interfaces using **Angular**, **React**, and **Material** libraries with focus on accessibility and offline support; optimized to reduce initial load times by **40%**.
 - Used **PostgreSQL**, **MySQL**, **MongoDB**, and **DynamoDB**; optimized indexes and queries for low-latency reads and configured replication for availability.
 - **AWS** Certified Developer – Associate; provisioned a variety of **AWS** services with **CloudFormation** for **IaC**.
 - Integrated **Datadog**, **ELK**, **Grafana**, and **LogRocket** for monitoring, debugging, and usage analytics.
 - Used **Docker** and **Kubernetes** for container builds, orchestration, and scaling.
 - Practiced **TDD** using **JUnit**, **Mockito**, **Jest**, and **Jasmine**. Strong advocate for **Agile**.
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EXPERIENCE

SIEMENS | SOFTWARE ENGINEER

March 2024 - Present | Remote

GENAI STACK - LANGCHAIN | GOOGLE ADK | RAG | EMBEDDING PIPELINES | PGVECTOR | MCP | A2A | AG-UI | PYDANTIC EVALS | LANGFUSE

- Took enterprise **Generative AI Solutions** from discovery and rapid prototypes through production rollout and feedback-led iteration.
- Built stateful **Multi-Agent Systems** with **LangChain** And **Google ADK**, covering tool routing, durable memory, model fallbacks, human approvals, and long-running workflows.
- Built **RAG And Embedding Pipelines** with semantic chunking, metadata enrichment, incremental indexing, hybrid retrieval, filtering, and reranking in **Pgvector**.
- Built secure **MCP Clients And Servers** for tools, resources, and prompts over Streamable HTTP using OAuth 2.1, protected-resource discovery, DCR, and PKCE.
- Implemented **A2A** Agent Cards and task workflows for agent discovery, delegation, multi-turn context, streamed status and artifacts, cancellation, and push notifications.
- Built **AG-UI** event streams for messages, tool calls, shared state, human approvals, interrupts, and resumable sessions in user-facing applications.
- Built **Pydantic Evals** quality gates with golden datasets, structured-output checks, retrieval and trajectory tests, LLM judges, and adversarial cases.
- Used **Langfuse** to trace model, retrieval, and tool execution and monitor tokens, first-token latency, cost, retrieval quality, failures, and user feedback.

EPAM SYSTEMS | SOFTWARE ENGINEER

Aug 2023 - March 2024 | Gurugram, Haryana

TECH STACK - JAVA | SPRINGBOOT | REACT | POSTGRESQL | AWS

- Developed an **Enterprise Service Bus (ESB)** with **Custom Data Transformation**, **Batch Processing**, and **Parallelization** to reduce exchange latency by **40%**.
- Built an **Asset Classification And Protection (ACP)** tool with **SOP Modeler**, enforced **Role-Based Access Control (RBAC)** and **Audit Logging** per asset and procedure.
- Implemented **Real-Time Alerts** and **Anomaly Detection** for unauthorized changes across physical and virtual assets.

DELOITTE | CONSULTANT

Jan 2022 - Aug 2023 | Gurugram, Haryana

TECH STACK - NESTJS | REACT | DYNAMODB | AWS

- Built claims processing dashboard for insurance ops, using **Machine Learning** to flag fraudulent claims.
- Used **AWS Step Functions** for orchestration across services, unified data access via **GraphQL**.
- Integrated **Kafka** with **WebSockets** for real-time call alerts and event-driven notifications.

TATA CONSULTANCY SERVICES | SYSTEMS ENGINEER

Aug 2019 - Jan 2022 | Indore, Madhya Pradesh

TECH STACK - JAVA | SPRINGBOOT | ANGULAR | MONGODB | AWS

- Created a dashboard for policyholders to view insurance details via a unified interface
- Built a **Camunda**-based workflow to issue **Letter Of Authority (LOA)** to authorized entities
- Migrated application from on-premise servers to **AWS**, enabling scalability and resilience

SKILLS

Generative AI & Agent Systems: LangChain, Google ADK, RAG, Embedding Pipelines, pgvector, MCP, A2A, AG-UI, Pydantic Evals, Langfuse

Frontend Technologies: JavaScript, TypeScript, Angular, React, HTML, CSS, Redux, NgRx, RxJS, Bootstrap, Angular Material, Material UI

Backend & APIs: Java, Spring Boot, Hibernate, Java EE, Node.js, NestJS, Express.js, GraphQL, PostgreSQL, MongoDB, DynamoDB, Redis

Cloud, DevOps & CI/CD: AWS (ECS, SQS, SNS, S3, CloudWatch, CloudFormation), Docker, Kubernetes, Git, Jenkins, Bamboo

Monitoring, Analytics & Messaging: Apache Kafka, WebSockets, Datadog, LogRocket, Splunk, Quantum Metric, ELK

CERTIFICATIONS

AWS CERTIFIED DEVELOPER - ASSOCIATE

EDUCATION

SRM INSTITUTE OF SCIENCE & TECHNOLOGY | BACHELOR OF TECHNOLOGY

July 2015 - May 2019 | Chennai, Tamil Nadu

GRADUATED WITH DISTINCTION